

ĐẠI HỌC QUỐC GIA HÀ NỘI
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NGHIÊN CỨU GIẢI PHÁP TỐI ƯU
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SOA and Related Works

1.1 Inertial Parameters Estimation

To the best of our knowledge, there has been no comprehensive review dedicated specifically to center-of-mass (CoM) estimation for manipulated objects. Existing surveys primarily focus on the identification of the complete set of inertial parameters rather than CoM estimation itself. A survey of Schedlinsk [1] presents a literature review of existing methodologies on how to estimate these parameters in mechanical engineering, whereas Mavrakis and Stolkin [2] is specific to robotics. According to Mavrakis and Stolkin's taxonomy, the methods for estimating inertial parameters can be categorized into three main groups by degree of interaction — purely visual, exploratory, and fixed-object.

The first visual methods relied on the assumption of a known and uniform object density in order to simplify the estimation problem by reducing it to the computation of the object's volume. In [3] the object is represented as a collection of equally sized voxels with identical masses, from which the center of mass is determined by computing the moment of inertia matrix. One of the most widely recognized approaches is presented in [4], the author decomposed a rigid body into polyhedral components with uniform density and computed the mass, and inertia tensor of each component before combining them to obtain the inertial properties of the complete object. The computation was achieved by converting the required three-dimensional volume integrals into two-dimensional surface integrals and one-dimensional line integrals using the divergence theorem, projection, and Green's theorem. However, while visual perception can accurately capture an object's shape and volume, the underlying density distribution is generally unknown, may vary throughout the object, and cannot be directly observed solely from visual information.

Exploratory and fixed-object methods, on the other hand, involve physical interaction with the object to perform the estimation task. These interactive techniques are commonly formulated based on the Newton–Euler equations of motion, which establish a mathematical relationship between the forces/torques (wrenches) applied to an object and its resulting motion. Properties like the center of mass (CoM), mass and the inertia tensor are treated as constant parameters within these equations and are subsequently estimated from the resulting system of equations, typically using least-squares techniques. In fixed object methods [5, 6, 7], the unknown object is rigidly connected to the robotic system, during the estimation process, either through a firm grasp or by being directly attached to the end-effector. Excitation trajectories are then executed by the robotic system to retrieve the necessary data from wrist sensors. The approach proposed in [8] employed a robotic arm to perform three distinct trajectories to distinguish the individual effects of mass, CoM, and the inertia tensor. The fundamental principle of this method can also be applied to different robotic platforms. In [9], the authors use a UAV equipped with two customised grippers to grasp the object and estimated the payload mass, CoM, and inertia from the UAV dynamics and measured motion, while [10] extended this concept to a hexacopter equipped with a 2-DOF manipulator, enabling simultaneous payload estimation and adaptive motion control. Although this method can achieve the most accurate full three-dimensional inertial properties under controlled conditions, its requirements for a secure grasp and specialized force/torque sensors makes it less versatile than exploratory methods in unconstrained or hazardous scenarios.

Exploratory methods exploit the measurements by utilizing more basic interaction such as such as poking, pushing or tilting. As mentioned, early studies of this category similarly estimates the unknown properties by using the model-based approach. Yu et al. [11] employed a two-fingered robot manipulator equipped with force sensors to estimate an object’s mass, 2D CoM, rotational inertia, and motion friction parameters from multiple pushing motions using a least-squares approach. For CoM estimation, they introduced C.M. lines, which are parallel to the object’s straight-motion direction and pass through its CoM. These lines were identified from fingertip force and motion measurements under different pushing directions, with their intersection determining the CoM. Franchi et al.[12] developed a decentralized approach for estimating an object’s mass, 2D CoM, and rotational inertia using multiple mobile robots. Information on the applied forces and contact geometry is shared among the robots through a consensus algorithm, enabling distributed

estimation of the unknown inertial properties. The method was subsequently extended in [13] to rely solely on noisy end-effector velocity measurements.

Another emerging line of research focuses on CoM estimation without requiring an explicit dynamic model. A number of studies have been built upon Mason’s Voting Theorem [14], which establishes a geometric relationship between the pushing direction and the resulting rotational motion of a planar object, and reducing the search space through successive bisection. By additionally incorporating modified binary search algorithm [15], Hyland et al. [16] proposed iterative pushing method using a holonomic mobile robot and external motion capture to derive the CoM location from multiple lines on which the CoM lies, called lines of action (LoA). This works is later extended in [17] into an infrastructure-free approach by utilizing onboard force and vision sensors. Uemura et al. [18] presented a dynamic control based method utilizing a nonlinear feedback control law to continuously adjusting the finger’s motion to suppress the object’s rotation, achieving pure translational motion along the identified line of action (LoA).

A growing body of work has explored data-driven approaches that learn CoM-related information directly from observed interaction data. Gao et al. [19] introduce a force-sensor-free exploratory framework that integrates high-speed robotic pushing with Mason’s Voting Theorem and recurrent neural networks to estimate the CoM of previously unseen objects with non-uniform density. In [20], the authors introduces a data-driven framework that estimates an object’s 2D inertial parameters by training a Multi-Output Regression Random Forest on physical features extracted from a single robotic push. By combining analytical pushing mechanics with machine learning, the method achieves robust and accurate localization across various surface types without the need for specialized hardware. These methods enable rapid estimation with fewer physical interactions. However, their reliance on prior training data may limit their generalization to previously unseen objects.

Methodology

2.1 Methodology

2.1.1 Formulation

The system consists of N omnidirectional mobile robots manipulating an object with polygonal shape whose motion is constrained to a 2D plane using pushing interaction. The primary objective is for the robots to cooperatively perform the estimation task to obtain the Center of Mass (COM) of the object in a distributed manner. It is assumed that the object's position and 2D geometry is known to all robot and neither the object nor the robots are subject to kinematic constraints. Each robot i has a circular footprint with radius r_R , can exert a normal force f_i within a limited range : $0 < f_i < f_{max}$, as well as measure the position p_i and velocity $\dot{p}_i$ at its contact point C_i on the object's surface. The forces applied to the object by the robots are represented by the vector: $\mathbf{f} = [f_1, \dots, f_N]^T \in \mathcal{F} := \{\mathbf{f} \in \mathbb{R}^N \mid 0 \leq f_i \leq f_{max}, i = 1, \dots, N\}$. It is also assumed that the robots are able to maintain continuous contact with the object and control an accurately regulated pushing force at its corresponding contact point throughout the manipulation process. The net force and torque that all robots exert on the object in the world-fix frame W are descibe as:

$$\mathbf{F} = \sum_{i=1}^N \mathbf{F}_i \quad (2.1)$$

$$\boldsymbol{\tau} = \sum_{i=1}^N (\mathbf{p}_i - \mathbf{p}_C)^{\perp T} \mathbf{F}_i \quad (2.2)$$

where $(\cdot)^{\perp}$ denotes the perpendicular operator, defined as $\mathbf{v}^{\perp} = Q\mathbf{v}$, with

$Q = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. $F_i = f_i^{\perp T} n_i$ is the force vector of robot i , with n_i being the unit normal vector at the contact point C_i . The friction model is negligible since we do not require a dynamically accurate model. The equation (2.2) can be rewritten as:

$$\tau = \sum_{i=1}^N \mathbf{p}_i^{\perp T} \mathbf{F}_i - \mathbf{p}_C^{\perp T} \mathbf{F} \quad (2.3)$$

2.1.2 Estimation algorithm

When the object undergoes rotational motion, all contact points rotate instantaneously about the same instantaneous center of rotation (ICR). The velocity of each contact point is tangential to its instantaneous circular trajectory and therefore perpendicular to the line connecting the contact point to the ICR. Consequently, the ICR can be geometrically determined from the velocity measurements at multiple contact points by the equation below:

$$\mathbf{p}_{ICR} = \mathbf{p}_i - \mathbf{p}_o + \frac{1}{\omega_o} (\dot{\mathbf{p}}_i)^{\perp} \quad (2.4)$$

where p_o and ω_o are the position and angular velocity of the object, respectively.

The proposed distributed estimation algorithm idea is to generate a pure rotational motion of the object by controlling formation and the exert force of the robots. When this occurs, the ICR is located exactly at the COM of the object. Note that since the net force F is independent of the object's COM position, when the object undergoes pure rotation, in other words, the net force F on the object is zero, then (2.3) reduces to:

$$\tau = \sum_{i=1}^N \mathbf{p}_i^{\perp T} \mathbf{F}_i \quad (2.5)$$

Therefore, the net torque τ is only dependent on the contact points position and the exerted forces, regardless of the object's COM position. By controlling the formation of the robots and their exerted forces, we can achieve the desired net torque τ_d that will induce a pure rotational motion of the object. This torque can be agreed upon by all robots through communication or a consensus algorithm. The COM position can then be estimated by each robot using the following equation:

$${}^B\mathbf{p}_C = R^T(\theta) \left[{}^W\mathbf{p}_i - {}^W\mathbf{p}_o + \frac{1}{\omega} ({}^W\dot{\mathbf{p}}_i)^\perp \right] \quad (2.6)$$

The estimation problem can now be formulated into 2 sub-problems: formation synthesis and force-based control. The formation synthesis problem's objective is to find a configuration p , which is able to cooperatively exert the desired force and torque on the object. The force control problem determine the forces to be exerted by each robot to achieve the desired net force and torque on the object.

2.1.3 Formation synthesis

To efficiently search for the optimal robot configuration, the search space is defined directly along the object boundary. Since each robot is constrained to establish contact with the object's surface, the position of robot i can be parameterized by a scalar variable s_i representing the arc length along the polygonal boundary of the object. Accordingly, the search space for (N) robots is expressed as $\mathbf{s} = [s_1, \dots, s_N]^T$, where $s_i \in [0, L)$ and (L) denotes the total perimeter of the object. To improve contact stability, a small margin d_s around each polygon vertex is omitted from the search space, the resulting feasible search space is illustrated in.

Since the maximum pushing force $f_{\max}$ is finite due to the physical limitations of the robots, for each configuration of the robots, the corresponding manipulation space is a closed set W consisting of all feasible set of forces and torques, or wrenches that can be exerted by the current formation. The full manipulation space can not be accurately determined or analyzed without prior information about the COM position, which heavily affects the rotational motion of the object. However as established above, when pure rotation occurs, the torque becomes decoupled from the COM position. The maximum and minimum torque can be exerted by a configuration p can be derived by solving the following linear programming problems:

$$\begin{aligned} \tau_{\max}(\mathbf{p}) &= \max_{\mathbf{f} \in \mathcal{F}} \sum_{i=1}^N \mathbf{p}_i^{\perp T} \mathbf{F}_i \quad \text{s.t.} \quad \sum_{i=1}^N \mathbf{F}_i = \mathbf{0}, \\ \tau_{\min}(\mathbf{p}) &= \min_{\mathbf{f} \in \mathcal{F}} \sum_{i=1}^N \mathbf{p}_i^{\perp T} \mathbf{F}_i \quad \text{s.t.} \quad \sum_{i=1}^N \mathbf{F}_i = \mathbf{0}. \end{aligned} \quad (2.7)$$

These two quantities are, in fact, the intersection of the manipulation space with the torque axis. A formation is considered feasible if the desired torque τ_d

is within the range of $[\tau_{\min}, \tau_{\max}]$. It is also beneficial to maximize the minimum distance between the desired torque τ_d and the boundary of the feasible torque range, which can be formulated as a fitness function J_r for robustness metric:

$$J_r = \min (|\tau_{\max} - \tau_d|, |\tau_{\min} - \tau_d|). \quad (2.8)$$

The formation synthesis problem is formulated as:

$$\min_{\mathbf{p} \in \mathbb{R}_{\geq 0}^N} J(\mathbf{p}) = -J_r(\mathbf{p}) \quad (2.9a)$$

$$\text{s.t.} \quad \exists \tilde{\mathbf{f}} \in \mathcal{F} \quad \text{s.t.} \quad F = \mathbf{0}, \quad \sum_{i=1}^N \mathbf{p}_i^{\perp T} \mathbf{F}_i = \tau_d, \quad (2.9b)$$

$$\text{frD}(p_i, p_j) \geq d_{\min}, \quad \forall i \neq j. \quad (2.9c)$$

where $d_{\min} > 2r_r$ is the a minimum Euclidean distance between the robots to avoid collision. It is essential that the constraints (2.9b) and (2.9c) are satisfied to ensure that the desired wrench can be exerted by the formation without violating physical feasibility, while minimizing the objective function yields a configuration that is optimal with respect to the selected criteria. The objective function (2.10) is a desirable property, which can be modified to include other metrics. In this case, only rotational robustness is considered. The formation synthesis problem is a non-convex optimization problem, which can be solved using heuristic approaches, for example Particle Swarm Optimization (PSO). To keep the distributed nature of the system, each robot can run the optimization algorithm independently and share their best particle with their neighbors. The best solution is then selected as the final configuration.

2.1.4 Force control

The next problem to be solved is to determine the forces to be exerted by each robot to achieve the desired net force and torque on the object. While one naive approach would be to just solve (2.1) and (2.3) using least-square approaches, this would fail to account for the physical limitations of the forces exerted by the robots, resulting in a solution that is not physically feasible. To tackle this issue,

the force control problem is formulated as an optimization problem:

$$\min_{\mathbf{f}} \quad \left(\sum_{i=1}^n \mathbf{p}_i^{\perp T} \mathbf{F}_i - \tau_d \right)^2 + \lambda \|\mathbf{f}\|_2^2, \quad \lambda > 0 \quad (2.10)$$

$$\text{s.t.} \quad \mathbf{F} = \mathbf{0}, \quad (2.11)$$

$$0 \leq f_i \leq f_{\max}, \quad i = 1, \dots, n. \quad (2.12)$$

Satisfying constraints (2.11) and (2.12) is the primary requirement of the optimization problem, since they ensure the physical feasibility and enable the desired pure rotational motion. It is desirable that the net torque is as close as possible to the desired torque τ_d , however it is not required to exactly match τ_d , provided that its magnitude is sufficient to overcome the resisting static friction torque. The second term in the objective function (2.10) is a regularization term that minimizes the total exerted force, which is beneficial for energy efficiency and to avoid saturation of the actuators. The optimization problem can be solved using standard convex optimization techniques, such as quadratic programming. Each robot can independently solve this optimization problem using its local measurements and shared information from its neighbors, ensuring a distributed implementation of the force control strategy.

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